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Upper-limb prosthetic requirements from the healthcare providers, end-users and relatives' perspectives

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ABSTRACT

Background: Upper-limb amputation negatively impacts people's quality of life because replicating the hand anatomy and its multiple functions is complex. This difficulty has caused prosthesis rejection rates to be as high as 50%, mainly due to a lack of function, comfort, and control.**Purpose:** This study aimed to define upper-limb prosthesis design requirements from the perspectives of health providers, end-users, and close relatives.**Methods:** Semistructured interviews were conducted with three types of stakeholders: 11 healthcare providers, 10 end-users with unilateral upper-limb amputation, and 10 close relatives. The questions included characteristics for prosthetic design and the capability to conduct activities of daily living (ADL) with different types of prostheses. This study forms part of a project focused on the design of a myoelectric upper limb prosthesis. The presented information pertains to the needs definition stage of the Design Thinking process, specifically, the Data, Information, Knowledge, and Wisdom scheme, aiming to outline the gathered insights and key findings descriptively.**Results:** The three groups agreed that the most critical characteristic of a prosthesis was comfort, followed by maintenance and low weight. Cost was the requirement with the most remarkable difference in perception among stakeholders. Differences in the perception of the capability of conducting activities of daily living (ie, transport, housework, personal care and grooming) were found between the end-users and their relatives. These results and previous research provide relevant information for device design, mainly related to critical challenges and user requirements. These considerations may help to improve user satisfaction and reduce device abandonment rates.**Conclusions:** All stakeholders prioritize comfort in upper-limb prostheses, though perceptions of cost diverge sharply depending on financial responsibility. Integrating users' ADL requirements could enhance independence. Tailoring designs to end-users' diverse needs, addressing family impacts, and mitigating external abandonment factors (eg, unrealistic expectations, training gaps, and accessory shortages) are critical to reducing device abandonment and improving rehabilitation outcomes.© 2025 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Introduction

Upper-limb amputation can limit the capability of performing social, professional, and other activities of daily living (ADLs). In comparison with different types, upper-limb amputation has a more considerable impact on people's quality of life because of the complexity of replicating the hand's anatomy and its multiple functions, starting with the several grasping possibilities, going from precision to gross grips.^{1–4} Consequently, high rejection rates are commonly seen among end-users when the prosthesis is not used in the first place (ie, primary rejection) or when the users start but later stop

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using the prosthesis (ie, secondary rejection). Rejection rates in upper-limb prostheses have values as high as 50% for mechanical, 53% for cosmetic, and 39% for myoelectric prostheses.⁵

The main reasons for primary rejection are the lack of perceived need (the end-user perceives activities can be easily performed without a prosthesis, like taking a bath) and the discrepancy between the demand and availability of prostheses complying with the user's needs in the market.⁶ Moreover, rejection has also been related to high weight, lack of comfort, limited functionality, high cost, limited use with specific clothing, skin irritation, increase in temperature at contact areas, low durability, low resistance, difficulty to control, esthetics, poor intrinsic user skills, and lack of haptic feedback, formal education, and prosthetic training.^{5,7–9} However, the main prosthetic requirements that continue to be a challenge are function, comfort, and control.^{5,6}

Several authors have stated that comprehensive research on upper-limb prostheses' design requirements should be focused on the future needs of users,³ comprising a broader scope beyond simple usefulness.¹⁰ Design requirements are defined as the attributes that enable the design team to convert ideas into features a device must comply with.¹¹ For this, Human Centered Design methodology involves users through the device design process: In the inspiration phase, to collect information about their needs; in the ideation phase, cocreating ideas that may solve the problem previously defined; and in the implementation phase, gathering feedback of the developed prototypes.¹² Moreover, the inclusion of the complex perspectives of the end-user in the design process has been proven to decrease abandonment and improve satisfaction.^{3,6,10,13,14}

In addition, previous studies indicate that the inclusion of stakeholders could help to understand better the context for which the device is developed, as well as the strategies for its implementation, and avoid potential barriers to its integration into the healthcare system.^{15,16}

Previous studies that included the views of upper-limb amputees have reported main design requirements, where the most important are comfort, durability, lightweight, and cost.^{5,14,17,18} Lightweight is the only characteristic reported by all authors and identified as a key requirement due to its effect on user perception.

It is essential to align these design requirements with performance in ADLs'. For example, tasks such as eating and dressing (common to all users, regardless of the level of limb loss or type of prosthesis) directly affect self-care and independence.³ Addressing these needs ensures an improvement in independence, reducing rejection by emphasizing real-world utility.

No studies were found that explored the points of view of other actors, such as health providers or relatives of people with amputation, on upper-limb prostheses. Most studies exploring prosthetic requirements do so from the end-users' point of view. Also, to the authors' knowledge, no study includes the analysis of ADLs that significantly impact user independence. Those ADLs can later be considered during the design process to implement related functionalities (eg, degrees of freedom). Therefore, there is a need to broaden the scope of these analyses by including various stakeholders in exploring preferences and priorities for prosthetic components and functions. Consequently, this study aimed to include the perspectives of health providers, end-users, and close relatives to define upper limb prosthesis design requirements.

Materials and methods

A descriptive study was performed to obtain information from end-users, their relatives (eg, caregivers), and healthcare providers to establish the prosthetic design requirements. Semistructured interviews were conducted with the three types of stakeholders, and three interviews were designed, one for each stakeholder (for more details on the raw data from the interviews, see [Annex 4](#)).

Previous literature indicates that data saturation is typically achieved after approximately 12 interviews,¹⁹ and focus groups are recommended to include up to 10 participants to maintain interactive depth.²⁰ Verbatim transcripts were analyzed in detail to prioritize deep insights into participants' experiences. Therefore, a sample size of 10 end users was selected.

Since one of the intentions was to enhance user independence with the prosthesis, a deeper understanding of ADL was needed. For this, a list of typical ADLs conducted with upper limbs (eg, self-care, transport, personal care, grooming, social activities) was defined based on the Brief Activity Performance Measure for Upper Limb Amputees (BAM-ULA) and the Opus Upper Extremity Functional Status (OPUS-UEFS).^{21–24} These surveys are used to evaluate the functionality of a person's upper body quantitatively; therefore, they were used as the basis for the considered activities to guarantee the inclusion of relevant ADLs previously validated by the literature.

The interviews included closed and open-ended questions and were created in Google Forms (Google, CA) to guide the researchers when conducting the interviews. All three interviews included items to evaluate the importance of prosthetic characteristics using a Likert scale. However, only end-users and their close relatives were asked to assess the capability of the end-users to conduct ADLs. Hence, the interviews contained a common section (prosthetic characteristics) for the three stakeholders and the prosthetic performance section that was excluded for healthcare providers. The healthcare providers were excluded from this domain as they do not have evidence of the end-user's daily use of the prosthesis.

End-users were included if they had unilateral transradial upper limb amputation and used any prosthesis. People with bilateral amputation were not included in the study due to the considerable difference in functionality between people with unilateral and bilateral amputation.³ There was no discrimination in the cause of the amputation, neither if it was congenital, traumatic, or nontraumatic. The study aimed to collect information from upper-limb prosthetic users, and as this group represents a minimum percentage of all upper-limb amputees, we wanted to have as many people in the study as possible. People with cognitive impairment were excluded from this study, as they likely could have difficulties following the interview sequence. Relatives were included if they lived with or spent at least 1 day a week with the end-user. Healthcare providers were included if they had experience in the amputation field (ie, physiatrists, physical therapists, ortho-prosthetists, orthopedic and hand surgeons).

End-users were recruited from databases of the prosthetic company involved in the present study and other databases obtained from prosthetic laboratories in the same city. When conducting the interview, the end-users were asked for their closest relative, and they were invited to participate. Healthcare providers were recruited from the previously mentioned prosthetic laboratories, and then the snowball method was used to increase the sample.

There was no sample size rationale as upper limb amputation is scarce worldwide, and only descriptive data were searched for the study's main objective.

One of the researchers contacted participants by phone. First, the objective was explained, and their oral consent was obtained. The interview was recorded, and the answers were registered in Google Forms software (Google, CA).

After the interview, the quantitative data on the prosthetic characteristics were organized using Excel (Microsoft, WA). The scheme DIKW (Data, Information, Knowledge, and Wisdom) from the Design Thinking approach was used to analyze the additional qualitative information obtained during the interview. This scheme starts with extracting participants' quotes (Data), interpreting this data (Information), clustering similar information and searching for patterns between participants (Knowledge), and finally, generalizing

these findings (Wisdom).²⁵ This four-step procedure was made using the MURAL platform (MURAL, CA) for better visualization (for more detail of the most critical verbatims of the three stakeholders, see Annexes 1, 2, and 3). The results from the interviews were later used to define design requirements. Translating this data into measurable prosthetic requirements is described in detail in a later article by the same authors.²⁶

This study was part of a project to design and evaluate the effectiveness of a myoelectric prosthesis (AKTIO). The Bioethics Committee of the School of Medicine of the Universidad de Antioquia approved the study protocol (approval record 012 – June 2019). All participants gave informed consent to participate in this study.

Results

The interviews were conducted with 11 healthcare providers, 10 subjects with unilateral upper-limb amputation, and 10 relatives (the raw data is available as a [Supplementary Data](#) file). Detailed interview quotes, originally in Spanish and translated into English, are available in the [Supplementary Data](#).

Table 1 presents the sociodemographic characteristics of study participants with unilateral upper-limb amputation, including age, cause, and time since amputation. Dominant and residual limb lengths were included because they can influence functional outcomes and the potential for prosthetic use.

Prosthetic characteristics

Table 2, presents the mean ratings and standard deviation of importance assigned by the stakeholder groups to the prosthetic characteristics, and the mean and standard deviation obtained among the three stakeholders.

Figure 1 is included to facilitate visualization of the data shown in **Table 2**, presents the average importance ratings assigned by stakeholder groups to prosthetic characteristics. The axes range was manipulated, starting at 3.5, to give a clearer view of the difference in opinions among stakeholders.

From **Table 2** and **Figure 1** can be noticed that comfort was the highest priority prosthetic requirement across all stakeholder groups. The most pronounced disparity occurred in the prioritization of cost, with healthcare professionals placing greater importance on this factor than end-users (8.5 vs 6.0). In addition, appearance demonstrated lower importance among healthcare professionals compared to end-users and family members (6.5 vs 7.7 vs 8.7,

Table 2

The importance of the characteristics according to the different groups

Characteristic	End-users	Relatives	Health staff	Average
Comfort	5 ± 0	5 ± 0	4.9 ± 0.3	5 ± 0.2
Socket fit	4.9 ± 0.3	4.8 ± 0.4	4.9 ± 0.3	4.9 ± 0.4
Low weight	4.8 ± 0.6	4.9 ± 0.3	4.8 ± 0.4	4.8 ± 0.5
Ease of access	4.6 ± 0.7	5 ± 0	4.9 ± 0.3	4.8 ± 0.5
Easy to don and doff	4.6 ± 0.5	5 ± 0	4.8 ± 0.4	4.8 ± 0.4
Maintenance	4.8 ± 0.4	4.7 ± 0.7	4.9 ± 0.3	4.8 ± 0.5
Precision	4.7 ± 0.5	4.7 ± 0.7	4.8 ± 0.4	4.7 ± 0.5
Training	4.6 ± 0.7	4.7 ± 0.7	4.8 ± 0.4	4.7 ± 0.6
Grip force	4.8 ± 0.4	4.3 ± 0.7	4.6 ± 0.7	4.6 ± 0.6
Easy cleaning	4.3 ± 0.8	4.9 ± 0.3	4.6 ± 0.7	4.6 ± 0.7
Appearance	4.4 ± 0.8	4.9 ± 0.3	4.3 ± 1.1	4.5 ± 0.8
Cost	3.8 ± 1.1	4.4 ± 0.9	5 ± 0	4.4 ± 1

The rating scale is from one to five, where one is not important and five is very important.

[Mean ± standard deviation].

Average: mean between the all participants.

respectively), while grip strength was more of a priority for end-users than for healthcare professionals (8.3 vs 7.4).

Contrastingly, end-users had differing viewpoints regarding appearance and functionality. Some of them gave more importance to appearance, expressing that they "would like it to be true to the real hand and comfortable" (user 4); they use the prosthesis "When I go out for medical appointments, to church or a restaurant," "to go out, to go to church, events, etc." (user 2). Nevertheless, other users gave more importance to functionality, saying that "as far as appearance, I do prefer functionality, if I could have both that would be wonderful, but if not, functionality" (user 9); they use the prosthesis "at all times, [the user] only takes it off when he is in the house and wants to rest" (a relative of user 10).

In addition to the characteristics mentioned in the questionnaire, end-users also referred to special features they would like to have in their prostheses. These included durability, waterproofness, and safer grips for transportation (eg, bus, motorcycle, car). These features were expressed in their own words: "The most important thing for me would be to be able to take the bus and feel more confident and stable because most of the time, I do not have company. I have no one to help me. What a drag to tell someone always to accompany me" (user 7), "a prosthesis for one is the replacement of a limb, and although one would like to do everything, sometimes one ends up being limited by its durability." (user 3), "I do not do any household chores with the prosthesis to protect it, especially to avoid not getting it wet" (user 9), "if it rains, you basically cannot go

Table 1

Users' sociodemographic characteristics and end-user/relative relationship

User	Sex	Age range (y)	Cause of amputation	Time since amputation (y)	Amputated limb	Dominant limb	Residual limb length	Relative
1	Female	31–40	Occupational accident	3	Left	No	Distal third	Son
2	Male	31–40	Congenital	Congenital amputation	Left	No	Distal third	Not interviewed
3	Female	> 50	Gunpowder	66	Left	No	Distal third	Daughter
4	Male	31–40	Occupational accident	3	Right	Yes	Middle third	Spouse
5	Male	> 50	Occupational accident	7	Right	Yes	Proximal third	Spouse
6	Female	20–30	Occupational accident	< 1	Right	No	Middle third	Father
7	Male	31–40	Electric shock	2	Right	Yes	Proximal third	Spouse
8	Female	20–30	Automobile accident	6	Right	Yes	Distal third	Father
9	Male	31–40	Occupational accident	3	Left	No	Distal third	Spouse and Cousin
10	Male	> 50	Occupational accident	5	Right	Yes	Proximal third	Spouse

Residual limb length:

Distal third: Lower portion (closer to the wrist end) of the forearm/upper limb.

Middle third: Mid-portion of the forearm/upper limb.

Proximal third: Upper portion (closer to the elbow or shoulder) of the forearm/upper limb.

Relative: Some participants had more than one relative who provided support. *Not interviewed* indicates that a relative was identified but did not participate or was unavailable for an interview.

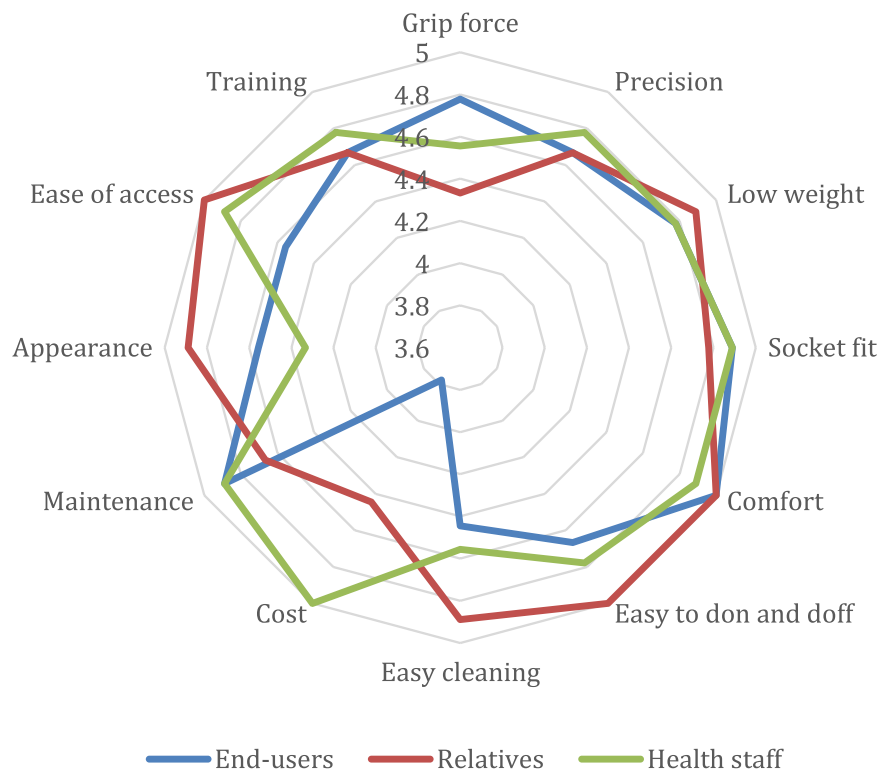


Fig. 1. The importance of the characteristics according to the different groups. The rating scale is from one to five, where one is not important and five is very important.

out" (user 9), "there would be activities in terms of my career where I could not use the prosthesis because it cannot get wet" (user 8).

Performance of ADLs

The perception of the activities' performance differed for end-users and their relatives. Table 2 shows end-users' and their relatives' perceptions of the tasks that the first ones can perform independently, can perform with the help of others, do not perform with the use of the prosthesis, or do not perform at all.

Table 3 shows the variation in the perception of the difficulty of carrying out the activities between users and their relatives (eg, for user 9, housework activities are challenging to perform, while according to the perception of his family member, the user can perform them easily).

The quotes of the subjects gathered during the interviews revealed that end-users and their relatives had different expectations about prosthesis performance and appearance when they first started to use it. They said that myoelectric prostheses "should have a little more movement, be less robotic and more like a real hand" (a relative of user 9), and they "thought that with the new prosthesis, they would be able to move all their fingers independently" (a relative of user 3). About the cosmetic prosthesis, they said that they "did not imagine they were going to find this kind of natural prosthesis" (a relative of user 6) and "wow! They were surprised because it was much better" (a relative of user 10).

Regarding the mismatch in expectations, one of the healthcare providers mentioned that "the patient's expectations must be clear. One should differentiate between what they say they want and what they really want" (healthcare provider 3). Furthermore, healthcare providers said that "the prosthesis will depend on the type of tasks" (healthcare provider 2), and because of this needed versatility for functioning based on the context, accessories might be required to adapt the prosthesis to its surroundings further ("Many of the

activities can be made with the help of accessories" (healthcare provider 1)). Also, due to the difficulty of replicating the hand functionality, one healthcare provider indicated that rehabilitation would focus on the preserved hand, and because of that, "the majority of activities will be made with the preserved hand, and in the other ones the prosthesis will be used as a support" (healthcare provider 2). The "other activities" mentioned refer to those requiring both upper limbs.

Both unmet expectations from end-users and relatives and topics referred by healthcare providers, such as the need for accessories and adequate rehabilitation, may be reasons for end-user abandonment, as they make the end-user feel uncomfortable with the prosthesis.

Additionally, since relatives may help the end-user to perform certain activities, they said they also experience economic and emotional changes after the amputation. Some of the quotes related to this topic were: "I help him a lot" (a relative of user 5), "anyway, the cost is high for us, but we do it for her" (a relative of user 6), "we [partners] are the ones who suffer the most from rejection" (a relative of user 9), "he would not tolerate anything, he wanted us to do things the way he expected. It made him feel desperate" (a relative of user 9).

Reasons for abandoning the device

In addition to the previously mentioned reasons for abandoning the device, the present study found the following: (1) unrealistic expectations from users and relatives regarding prosthesis functionality (eg, "it should have a little more movement, less robotic and more like a real hand" (relative 9)); (2) insufficient occupational therapy training on limb dominance and ADL (eg, "without good training, the patient will achieve nothing" (healthcare provider 1)); and (3) limited availability of specialized accessories for ADL

Table 3

Comparison between users and their relatives in the perception of their ability to perform tasks and the difficulty involved

ACTIVITY	1		2		3		4		5		6		7		8		9	
	U	R	U	R	U	R	U	R	U	R	U	R	U	R	U	R	U	R
Transport																		
Bus																		
Car																		
Motorcycle																		
Bicycle																		
Train																		
Taxi / Taxi apps																		
Walk																		
Housework																		
Dishwashing																		
Sweep																		
Mop																		
Laundry																		
Cook																		
Take care of children																		
Use the cell phone																		
Occupation																		
Study																		
Office work																		
Heavy work (with machinery, heavy tools, field)																		
Outdoor activities																		
Sport																		
Social activities																		
Family activities																		
Personal care and grooming																		
Bath																		
Get dressed																		
Brush the teeth																		
Go to the bathroom																		
Makeup																		
Shave																		
Eat																		

Table Notes

- U: user, R: relative

The user can perform it on their own with ease	The user can easily perform it accompanied
Users can perform it on their own with difficulty	The user can do it accompanied with difficulty
The user does not perform this task, or does not use the prosthesis.	

performance (eg, “many activities can be done with the help of accessories” (healthcare provider 1)).

Discussion

This study compared the perspectives of prosthetic users, their relatives, and healthcare providers on the importance of the characteristics of upper limb prostheses and the use of these devices in

performing ADLs. The main finding was that the most critical aspect of a prosthesis for the three groups was comfort. The prosthesis cost was the characteristic in which the three stakeholders' importance perceptions differed significantly. Most of the participants were young males, and the amputations were due to work-related trauma, which is in line with the published epidemiology of upper limb amputations.²⁷

Although there has been growing evidence that supports the participation of end-users in the design and prostheses prescription

process,^{3,5,6,10,13,14,17,18,28} only one study that compared the perspectives of different stakeholders was found.¹⁶ This latter study was conducted in lower limb prostheses and recommended incorporating both user and healthcare provider's perspectives to understand the prosthesis prescription process.²⁸ Hence, to the authors' knowledge, this will be the first article focusing on upper-limb prostheses.

This study also takes a novel approach by introducing the relatives' perspectives when exploring prostheses' requirements. The inclusion of other stakeholders in the design process offered valuable insights, such as the clarity emphasized by health care staff, indicating that the prosthesis must not intend to replace the residual limb but to provide support in bimanual activities or the need for the device to be waterproof expressed by the users, as stated in the previous article on the development of an assistive device.²³

According to the literature, reasons for prostheses' abandonment could be categorized into comfort and function.²⁹ On one hand, weight, temperature, and perspiration have been among the most common and persistent comfort-related reasons for abandonment. On the other hand, control and sensory feedback have been key concerns regarding function. In the present study, comfort, socket fit, and low weight were the most essential requirements for end-users, which is in line with other authors.^{5,9,14,17,18,30} Other characteristics related to comfort were ease of donning and doffing and ease of cleaning and maintenance, all of which were highly rated by end-users and their relatives.

In the published literature, function has been considered more important than appearance.^{14,18} In the present study, appearance was less critical to end-users than relatives. This is probably due to the users' pressing need for independence during their daily activities, contrasting with the relatives' lack of a more empathetic perspective. However, because end-users have a higher relevance when defining the prosthetic requirements, their opinions would have a higher weight in the final score (for more detail, see the article by the same authors²⁶). Their personality and lifestyle may influence the importance of these two characteristics for each user. Two types of users were found when considering the subjects' quotes gathered during the interview. The first preferred esthetic appearance and gave the prosthesis a social function (three users). The latter gave more importance to functionality and mainly used the prosthesis for the grip function (three users). The other users did not prefer these two characteristics (appearance and functionality). These users must be approached differently when designing or prescribing upper-limb prostheses. Cosmetic prostheses may be recommended for the first type of users, while myoelectric or mechanical prostheses would be the best option for the second type. The latter finding also suggests that upper limb amputees may need different types of prostheses to perform ADLs due to the hand complexity.^{1–4}

Also, in this study, the cost had the most considerable difference in perception depending on the viewpoint, being the least important requirement to end-users, and was only recognized as important by healthcare providers. However, for Biddis et al, the cost was the third most crucial concern to end-users.⁵ These differences may be related to the prosthetic financing process in each country and the several reimbursement policies applied by insurance companies and public healthcare systems.³ The authors speculate that in Colombia, the providers who prescribe the prostheses are limited by the payers or the insurers, which determine the components' selection and give more importance to the costs. Furthermore, end-users give less importance to cost as the healthcare system provides the prosthesis, and they do not have to self-finance any portion of the total cost.

When comparing the perception of users' performance in ADL using the prosthesis between end-users and their relatives, differences were found in actions such as cooking (ie, end-users believe that they cannot do it, and relatives believe that end-users can do it without help), mopping (ie, end-users believe they can do it without help and relatives believe that users cannot do it), and take the train

(ie, end-users believe that they cannot do it and relatives have mixed opinions, divided within the three options). For most activities, end-user opinions were divided between the impossibility of performing the activity and the capacity of doing it without help. The previously mentioned resembled the results from Biddis et al, in which 16% of the 178 individuals encountered no impossible challenges to overcome daily. In comparison, 10% of individuals found most activities, mainly bilateral tasks, challenging.⁵ Most of the categories of personal care and grooming activities were perceived by end-users and relatives as being easily performed. According to the systematic review by Cordella et al, the ADLs that the patients desired to perform with the prostheses were eating, dressing, typewriting, handling a cell phone, and opening a door.³ In the present study, end-users and relatives perceived that riding a bike and doing heavy work could not be performed by end-users. In the survey by Biddis et al, the most challenging activities reported by participants included household chores (eg, car repairs, gardening, doing dishes, cleaning, using tools, climbing stairs), playing sports (eg, cycling, swimming, ball sports), and performing hobbies (eg, playing a musical instrument, woodworking, and crafts).⁵ The most common difficulties in ADL experienced by amputees were lacing shoes, removing bottle tops with a bottle opener, and using scissors.³¹

Reasons for abandonment were found that are not directly related to the prosthesis:

- the unrealistic expectations of end-users and their relatives regarding prosthesis performance
- the fact that occupational therapy may not include sufficient training on limb dominance and ADL
- the scarcity of specific accessories required for the performance of ADL and other activities

The medical staff involved in the adaptation could address the first and second reasons, while the second and third reasons could be addressed by research or product development projects. This highlights the importance of including stakeholders in the rehabilitation process.

In addition, the quotes obtained from family members provide insight into the consequences they may suffer due to the end-user's amputation. They are possibly related to the added economic, physical, and emotional burden that comes with their relative's mobility limitation. Since this was not found in previous literature, subsequent studies are recommended to explore both the influence of family members in the rehabilitation process and the consequences they may suffer as caregivers.

Study limitations

Due to COVID-19 restrictions, the planned focus groups were replaced by telephone interviews. This change did not allow discussions between healthcare professionals, end users, and family members to be presented and for responses to be compared between participant groups. In addition, because it was conducted via phone call, limiting the ratings that could be assigned to the importance of prosthesis features was challenging. Therefore, stakeholders could assign the highest ratings (with the most results being approximately 5).

No occupational therapist was included among the interviewees. However, an occupational therapist is part of this study's team and contributed to the elaboration of the interview and the analysis of the results. Hence, she was not among the people interviewed.

Future recommendations

This study's findings can guide prosthetic device design, improving challenging areas and satisfying user requirements. Future

research should continue to explore the perceptions of end-users, their relatives, and healthcare providers on a larger scale to understand better the factors that explain prosthesis rejection.

Conclusion

Although stakeholders' perceptions of the importance of the characteristics differ, all three agree that comfort is the primary concern, underlining the need for designs to prioritize it. The most significant difference was cost, likely relevant to whoever must assume it or deal with its limitations. Also, considering ADLs with a more substantial impact on independence within the requirements could lead to a prosthesis that will further help the person with an upper-limb amputation.

In addition to the prosthesis, other factors should be considered for a successful fitting process, including the end-user's expectations of the device, the impact on family members, and an appropriate training process.

It is expected that considering the above in future developments of assistive devices will reduce abandonment due to design factors. It is important to highlight that factors external to the prosthesis were found to lead to device abandonment (eg, unrealistic expectations, insufficient training, and accessory shortages) and should be considered for a successful rehabilitation process.

Ethics approval and consent to participate

This study was part of a project to design and evaluate the effectiveness of a myoelectric prosthesis (Clinical Trials Registry NCT05510193). The Bioethics Committee of the School of Medicine of the *Universidad de Antioquia* approved the study protocol (approval record 012 – June 2019). All participants gave informed consent to participate in this study.

Author contributions

Sofía C. Henao: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Simón Cuartas-Escobar:** Writing – review & editing, Writing – original draft, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Sara Salazar-Salgado:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ana María Posada-Borrero:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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Data availability

The information obtained from the surveys is displayed on the graphs, while the verbatims were translated from Spanish to English and they are available for download as an attachment.

Declaration of Competing Interest

Sara Salazar-Salgado works in Prótesis Avanzadas S.A.S. which is a Colombian company that designs and manufactures myoelectric

upper-limb prostheses. The remaining authors have no competing interests.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jht.2025.04.006.

References

- Behrend C, Reizner W, Marchessault JA, Hammert WC. Update on advances in upper extremity prosthetics. *J Hand Surg.* 2011;36(10):1711–1717. <https://doi.org/10.1016/j.jhsa.2011.07.024>
- Cancio JM, Ikeda AJ, Barnicott SL, Childers WL, Alderete JF, Goff BJ. Upper extremity amputation and prosthetics care across the active duty military and veteran populations. *Phys Med Rehabil Clin N Am.* 2019;30(1):73–87. <https://doi.org/10.1016/j.pmr.2018.08.011>
- Cordella F, Ciano AL, Sacchetti R, et al. Literature review on needs of upper limb prosthesis users. *Front Neurosci.* 2016;10:1–14. <https://doi.org/10.3389/fnins.2016.00209>
- B. Rosenkrantz, S. Drucker, S. Salauddin, B. Lowe, Systems and methods for integrated robust orthotics and prosthetics hyper-customization, US20170285618A1, 5 de octubre de 2017. Accedido: 4 de agosto de 2021. [En línea]. Disponible en: (<https://patents.google.com/patent/US20170285618/en>).
- Biddiss E, Beaton D, Chau T. Consumer design priorities for upper limb prosthetics. *Disabil Rehabil Assist Technol.* 2007;2(6):346–357. <https://doi.org/10.1080/17483100701714733>
- Østlie K, Lesjø IM, Franklin RJ, Garfelt B, Skjeldal OH, Magnus P. Prosthesis rejection in acquired major upper-limb amputees: a population-based survey. *Disabil Rehabil Assist Technol.* 2012;7(4):294–303. <https://doi.org/10.3109/17483107.2011.635405>
- Dabaghi-Richerand A, Haces-García F, Capdevila-Leonori R. Prognostic factors of a satisfactory functional result in patients with unilateral amputations of the upper limb above the wrist that use an upper limb prosthesis. *Rev Esp Cir Ortop Traumatol.* 2015;59(5):343–347. <https://doi.org/10.1016/j.recot.2014.11.001>
- Duong T, Wagner B, Abraham T, et al. Comparative study of functional grasp and efficiency between a 3D-printed and commercial myoelectric transradial prosthesis using able-bodied subjects: a pilot study. *JPO J Prosthet Orthot.* 2017;29(3):112–118. <https://doi.org/10.1097/JPO.0000000000000130>
- Resnik L. Development and testing of new upper-limb prosthetic devices: research designs for usability testing. *J Rehabil Res Dev.* 2011;48(6):697–706. <https://doi.org/10.1682/jrrd.2010.03.0050>
- Kyberd PJ, Hill W. Survey of upper limb prosthesis users in Sweden, the United Kingdom and Canada. *Prosthet Orthot Int.* 2011;35(2):234–241. <https://doi.org/10.1177/0309364611409099>
- Privitera MB. Chapter 6 - Developing insights. In: Privitera MB, ed. *Contextual Inquiry for Medical Device Design*. Amsterdam, Netherlands: Academic Press; 2015:141–161. <https://doi.org/10.1016/B978-0-12-801852-1.00006-X>
- Cross N. *Engineering Design Methods: Strategies for Product Design*. Chichester, England; Hoboken, NJ: Wiley; 2008.
- Kerver N, van Twillert S, Maas B, van der Sluis CK. User-relevant factors determining prosthesis choice in persons with major unilateral upper limb defects: a meta-synthesis of qualitative literature and focus group results. *PLoS One.* 2020;15(6):e0234342. <https://doi.org/10.1371/journal.pone.0234342>
- Walker MJ, Goddard E, Stephens-Fripp B, Alici G. Towards including end-users in the design of prosthetic hands: ethical analysis of a survey of Australians with upper-limb difference. *Sci Eng Ethics.* 2020;26(2):981–1007. <https://doi.org/10.1007/s11948-019-00168-2>
- Pahl G, Beitz W, Feldhusen J, Grote K-H. *Engineering Design: A Systematic Approach*. 3rd ed London: Springer-Verlag; 2007. <https://doi.org/10.1007/978-1-84628-319-2>
- Franco-Trigo L, Fernandez-Llamos F, Martínez-Martínez F, Benrimoj SI, Sabater-Hernández D. Stakeholder analysis in health innovation planning processes: a systematic scoping review. *Health Policy.* 2020;124(10):1083–1099. <https://doi.org/10.1016/j.healthpol.2020.06.012>

17. Benz HL, Jia Y, Rose L, et al. Upper extremity prosthesis user perspectives on unmet needs and innovative technology. *Annu Int*. 2016;2016:287–290. <https://doi.org/10.1109/EMBC.2016.7590696>
18. Stephens-Fripp B, Jean Walker M, Goddard E, Alici G. A survey on what Australians with upper limb difference want in a prosthesis: justification for using soft robotics and additive manufacturing for customized prosthetic hands. *Disabil Rehabil Assist Technol*. 2020;15(3):342–349. <https://doi.org/10.1080/17483107.2019.1580777>
19. Guest G, Bunce A, Johnson L. How many interviews are enough?: An experiment with data saturation and variability. *Field Methods*. 2006;18(1):59–82. <https://doi.org/10.1177/1525822x05279903>
20. Krueger RA, Casey MA. *Focus Groups: A Practical Guide for Applied Research*. Thousand Oaks, CA: SAGE Publications; 2008.
21. Heinemann AW, Bode RK, O'Reilly C. Development and measurement properties of the Orthotics and Prosthetics Users' Survey (OPUS): a comprehensive set of clinical outcome instruments. *Prosthet Orthot Int*. 2003;27(3):191–206. <https://doi.org/10.1080/03093640308726682>
22. Burger H, Franchignoni F, Heinemann AW, Kotnik S, Giordano A. Validation of the orthotics and prosthetics user survey upper extremity functional status module in people with unilateral upper limb amputation. *J Rehabil Med*. 2008;40(5):393–399. <https://doi.org/10.2340/16501977-0183>
23. Jarl GM, Heinemann AW, Norling Hermansson LM. Validity evidence for a modified version of the Orthotics and Prosthetics Users' Survey. *Disabil Rehabil Assist Technol*. 2012;7(6):469–478. <https://doi.org/10.3109/17483107.2012.667196>
24. Resnik L, Borgia M, Silver B, Cancio J. Systematic review of measures of impairment and activity limitation for persons with upper limb trauma and amputation. *Arch Phys Med Rehabil*. 2017;98(9):1863–1892.e14. <https://doi.org/10.1016/j.apmr.2017.01.015>
25. Sanders I, Stappers PJ. *Convivial Toolbox: Generative Research for the Front End of Design*. Illustrated edition Amsterdam, The Netherlands: Laurence King Publishing; 2013.
26. Salazar-Salgado S, Escobar-Saltaren D, Cuartas-Escobar S, Calderón LA, Posada-Borrero AM, Henao SC. Complementing cross methodology with human centered design approach: application in the development of an assistive device. *Int J Ind Ergon*. 2025;105:103674. <https://doi.org/10.1016/j.ergon.2024.103674>
27. Ziegler-Graham K, MacKenzie EJ, Ephraim PL, Travison TG, Brookmeyer R. Estimating the prevalence of limb loss in the United States: 2005 to 2050. *Arch Phys Med Rehabil*. 2008;89(3):422–429. <https://doi.org/10.1016/j.apmr.2007.11.005>
28. Schaffalitzky E, Gallagher P, MacLachlan M, Wegener ST. Developing consensus on important factors associated with lower limb prosthetic prescription and use. *Disabil Rehabil*. 2012;34(24):2085–2094. <https://doi.org/10.3109/09638288.2012.671885>
29. Smail LC, Neal C, Wilkins C, Packham TL. Comfort and function remain key factors in upper limb prosthetic abandonment: findings of a scoping review. *Disabil Rehabil Assist Technol*. 2020;16(8):1–10. <https://doi.org/10.1080/17483107.2020.1738567>
30. McFarland LV, Hubbard Winkler SL, Heinemann AW, Jones M, Esquenazi A. Unilateral upper-limb loss: satisfaction and prosthetic-device use in veterans and servicemembers from Vietnam and OIF/OEF conflicts. *J Rehabil Res Dev*. 2010;47(4):299–316. <https://doi.org/10.1682/jrrd.2009.03.0027>
31. Jang CH, Yang HS, Yang HE, et al. A survey on activities of daily living and occupations of upper extremity amputees. *Ann Rehabil Med*. 2011;35(6):907–921. <https://doi.org/10.5535/arm.2011.35.6.907>

JHT Read for Credit

Quiz: # C30

Record your answers on the Return Answer Form found on the tear-out coupon at the back of this issue. There is only one best answer for each question.

- #1. It is felt that as many as ____% of upper limb prosthetic users end up not using their device
- 10
 - 25
 - 50
 - 75
- #2. The study design was
- qualitative
 - RCTs
 - case series
 - n = 1
- #3. Not included in the study were patients with_____ amputations
- traumatic
 - congenital
 - transradial
 - bilateral
- #4. The primary factor affecting acceptance was
- cost
 - appearance
 - comfort
 - function
- #5. The authors recommend a user-specific approach in addressing the needs of each patient
- not true
 - true